1 Solution — GIAM 1.2 Problem 7

1.1 Problem

Use the second definition of "prime" to see that 6 is not a prime. In other words, find two numbers (the a and b that appear in the definition) such that 6 is not a factor of either, but *is* a factor of their product.

1.2 The Second Definition

Recall the definition from Section 1.2:

> *A prime is a quantity p such that whenever p is a factor of some product ab, then either p is a factor of a or p is a factor of b.*

In divisibility notation: p is prime iff

$$p \mid ab \implies p \mid a \text{ or } p \mid b.$$

To show 6 is *not* prime, we just need *one* product ab that 6 divides while dividing neither a nor b.

1.3 The Counterexample

Take

$$a = 2, \qquad b = 3, \qquad ab = 6.$$

Then:

- $6 \mid 6$ — yes, 6 is a factor of the product $ab = 6$.
- $6 \nmid 2$ — 6 is *not* a factor of $a = 2$.
- $6 \nmid 3$ — 6 is *not* a factor of $b = 3$.

So 6 divides the product ab but divides *neither* a nor b. This directly violates the second definition, so 6 is **not prime**. ■

1.4 Why It Works

The reason 6 can be "cheated" this way is that $6 = 2 \cdot 3$ is *composite*: it is built from two different prime factors, 2 and 3. We hand the factor 2 to a and the factor 3 to b. Then:

- the product ab contains *both* prime factors, so $6 \mid ab$;
- but a contains only the 2 (missing the 3) and b contains only the 3 (missing the 2), so 6 divides *neither*.

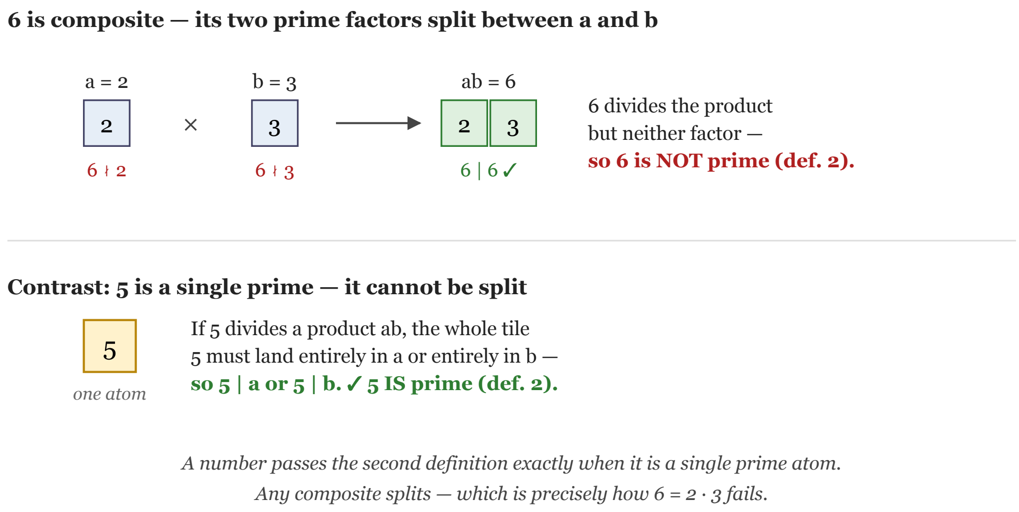

Figure 1.1: Top: $6 = 2{\cdot}3$ deals its factor 2 to a and factor 3 to b. The product has both $(6 \mid 6)$, but neither a nor b alone is divisible by 6. Bottom: a genuine prime like 5 is a single 'atom' that cannot be split — it must land wholly in a or wholly in b.

1.5 The General Principle

This is not special to 6. **Every** composite number n fails the second definition the same way. If n is composite, then $n = de$ with $1 < d < n$ and $1 < e < n$. Take $a = d$ and $b = e$:

$$n \mid ab = de = n \quad \text{(trivially)}, \qquad \text{but} \qquad n \nmid d \text{ and } n \nmid e,$$

since d and e are both strictly smaller than n (and positive), so neither can have n as a factor. Hence every composite violates the definition — exactly as it should.

For 6 the factorization is $6 = 2 \cdot 3$, giving the $a = 2$, $b = 3$ above.

1.6 Remark — More Counterexamples for 6

The product need not equal 6 exactly. Any a, b that split 6's prime factors 2 and 3 between them will do:

a	b	ab	$6 \mid ab$?	$6 \mid a$?	$6 \mid b$?
2	3	6	yes	no	no
3	4	12	yes	no	no
2	9	18	yes	no	no
4	9	36	yes	no	no
10	15	150	yes	no	no

Table 1.1

Each row works because one number supplies a factor of 2 (but no 3) and the other supplies a factor of 3 (but no 2).

1.7 Remark — The Two Definitions Agree for Integers

Section 1.2 gives two definitions of “prime”:

1. **(irreducible)** $p > 1$ whose only positive factors are 1 and p;
2. **(Euclid’s lemma)** $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.

For the integers these are *equivalent* — they pick out exactly the same numbers. We just saw one half of the equivalence (composite $\Rightarrow$ fails def. 2, equivalently def. 2 $\Rightarrow$ not composite). The other half — that a genuine prime p *does* satisfy def. 2 — is precisely **Euclid’s lemma**, the cornerstone fact that makes prime factorization *unique* (the

Fundamental Theorem of Arithmetic, used throughout Problems 1 and 4). The proof of Euclid's lemma uses the gcd / Bézout identity and is a highlight of any first number-theory course.

1.8 Remark — Why the Textbook Bothers With a Second Definition

If the two definitions agree for integers, why state both? Because they **stop agreeing in more general number systems**, and the second one is the "right" generalization. In ring theory the two notions get different names:

- an **irreducible** element cannot be factored into two non-units (definition 1);
- a **prime** element satisfies Euclid's lemma, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$ (definition 2).

Every prime element is irreducible, but *not* conversely. The classic example is the ring $\mathbb{Z}[\sqrt{-5}] = \{m + n\sqrt{-5} : m, n \in \mathbb{Z}\}$, where 2 is irreducible yet **not** prime — for exactly the reason 6 failed here:

$$2 \;\mid\; 6 \;=\; (1+\sqrt{-5})(1-\sqrt{-5}), \qquad \text{yet} \qquad 2 \nmid (1+\sqrt{-5}) \text{ and } 2 \nmid (1-\sqrt{-5}).$$

So 2 divides the product but neither factor — the *identical* splitting trick. This is why $\mathbb{Z}[\sqrt{-5}]$ has non-unique factorization ($6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$, two genuinely different factorizations — the same example flagged in Problem 1). The second definition is the one that survives into this setting and correctly distinguishes "true primes" from mere irreducibles.

1.9 Remark — Does 1 Satisfy the Second Definition?

The textbook posed this as an exercise, and the answer sharpens our understanding. Check definition 2 for $p = 1$: whenever $1 \mid ab$ (which is *always* true), is $1 \mid a$ or $1 \mid b$? Yes — 1 divides *everything.* So 1 satisfies the second definition *vacuously.*

This is exactly why the second definition, by itself, is *not enough* — it would wrongly admit 1 as a prime. The fix is to additionally require that p be greater than 1 (in ring-theoretic language, that p is *not a unit* and *not zero*). With that proviso, the two definitions agree perfectly for the integers. So 1 is excluded not because it fails Euclid's lemma — it passes — but by the separate ">1" clause, the same clause needed to keep the Fundamental Theorem of Arithmetic clean (Problem 2's remark).

1.10 Remark — Composite $\Longleftrightarrow$ Splittable

Pulling the threads together, we have a clean three-way equivalence for an integer $n > 1$:

$$n \text{ is composite} \iff n = ab \text{ with } n \nmid a,\ n \nmid b \iff n \text{ fails Euclid's lemma.}$$

The middle statement is the "splittability" we exploited; reading the chain right-to-left says that *passing* Euclid's lemma (plus being > 1) is the same as being prime. The exercise asked us to witness the leftmost-to-middle direction for the specific composite $n = 6$, and $a = 2$, $b = 3$ is the witness.